

Hugo BUSOLINI
Timothée CORDOBA
Alexi GRANDPERRET
Julie HASEGAWA

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Electric Production and Hydrogen

STUDY OF THE MULTI-DOMAIN IMPACT OF VARIOUS DILUTION RATIO OF SAF ON COMMERCIAL AIRCRAFT



Charles RIGOUDY
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First and foremost, we would like to thank Mr. Thomas Lesaffre, a PHD combustion research engineer, for his willingness to share his expertise on the implementation of SAFs and to guide us through the technical aspects of such a multidisciplinary topic. Although the academic needs of our study only led to a brief exchange, the success and the documentary foundation of our study nevertheless depended heavily on it. Thank you very much for your guidance and detailed insight on how to approach such a problematic.

In our 2nd year of aerospace engineering at IPSA, the *Electric Production & Hydrogen* course presented us with a list of numerous subjects tackling renewable propulsion technology challenges, including hydrogen usage and containment, electric technologies as well as SAFs. Being more interested in the latter, we chose a Sustainable Aviation Fuel related problematic for this subject applied investigation study.

As we deepened our researches, the emerging multidomain aspect of the problematic (originally oriented towards engine impacts of SAF usage) pushed us to redefine it, adopting a more multi-factor overview of the challenges linked to SAF usage. Our core problematic therefore reads:

Study of the multi-domain impact of various dilution ratio of SAF on commercial aircraft.

The present document overviews our investigation assessing the above subject, by first presenting the reader with a state-of-the-art context recalling before diving into the technical approaches chosen to tackle and illustrate the key points behind this problematic. A following non-technical analysis will further complete our investigation by adding an ecological, economical and geopolitical overview of the subject. In the end, we present to the theoretical client tasking us to study such problematic our synthesized response, summarizing all of the key points identified throughout the entirety of our study.

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I – State of the Art

A – Context and main objectives

As we navigate a century of unprecedented environmental challenges, the transition toward sustainable energy is no longer a matter of choice, but a real shift in how we conceive global mobility and its impact on the planet. Indeed, according to the **International Air Transport Association (IATA)**, the aviation sector represents **2 to 3% of the global CO2 emission** [1]. This figure might not seem significant but is actually alarming, as only a small part of the population is responsible for that number. Plus, aviation is one of the **fastest growing sectors** in terms of carbon footprint, with a growth rate that is outpacing many other industries. Therefore, it is a difficult sector to decarbonize. Another key point is that CO2 emissions are not the only parameter responsible for the aviation's environmental impact, **contrails** and **Nox emissions** also represent considerable contributions [2]. These non-CO2 emissions are estimated to increase the industry's total climate impact by a factor of two to three compared to CO2 alone.

Facing this climate emergency, international organizations had to come up with solutions. They established the **Paris Agreement** in 2015, aiming to limit global warming to **well below 2°C** and preferably below 1.5°C [3]. In reality, the international aviation was not explicitly detailed in the agreement as it is handled by the International Civil Aviation Organization (ICAO). However, the aviation industry had to align for the country to respect their engagements. As a result, organizations such as IATA and ICAO committed in 2021 to a strategic goal of '**Net Zero Carbon Emissions by 2050**' [4]. This very ambitious goal implies that any residual CO2 emitted by aircraft must be balanced by an equivalent amount of carbon removal from the atmosphere. In order to achieve Net Zero, it requires technological innovations and carbon offsetting as illustrated on the Appendix 1.

Several technological solutions such as **electric propulsion** and **hydrogen-powered aircraft** are being explored to reach this goal. However, these solutions face significant difficulties for long-haul aviation. On the one hand, for battery, the energy density remains too low for large aircraft. On the other hand, hydrogen requires a complete redesign of airframes and airport structures, which won't be ready until several decades.

Contribution to achieving Net Zero Carbon in 2050

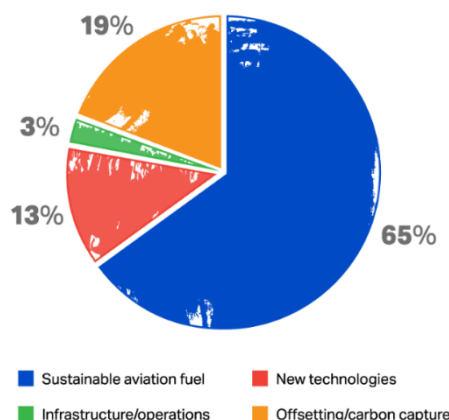


Figure A - SAF as a main contribution to achieving Net Zero Carbon in 2050

Therefore, **Sustainable Aviation Fuels (SAF)** emerge as the most relevant and **immediate solution**. Contrary to hydrogen or electricity, SAF is a '**drop-in**' fuel, which means that it is fully compatible with existing jet engines without requiring any aircraft modifications.

To accelerate this transition, the European Union has introduced the **ReFuel Aviation regulation** [5]. This compels fuel suppliers to incorporate increasing amounts of SAF into kerosene, starting with 2% in 2025, 6% in 2030, 20% in 2035 and aiming **70% in 2050**. In addition, the chosen SAF must be **sustainable** like advanced biofuels or synthetic fuels. The mandate explicitly excludes first-generation biofuels from food crops to avoid deforestation which would be counterproductive. This legal framework makes SAF not just a technical choice, but a **regulatory obligation for the future of the aviation industry**.

B – SAF, a new alternative

Now that we have reviewed the different possibilities for aviation to start pursuing the Net Zero goal, let's go in depth into the **most relevant solution: SAF**. Sustainable aviation fuels are a class of alternative fuels designed to **reduce the climate impact of aviation** while remaining **compatible** with the **current aviation** system. They are generally produced from renewable or waste-based feedstocks, such as used cooking oil, agricultural residues, municipal waste, or other biomass sources. These materials are chosen because they are either **non-fossil** in origin or would **otherwise be discarded**.

The potential to deliver substantial reductions in lifecycle greenhouse-gas emissions is very present. Unlike simple tailpipe comparison, lifecycle emissions consider the **full chain of fuel production**, including feedstock collection, transportation, processing, refining, distribution and final combustion in aircraft. Since many SAF feedstocks are renewable or derived from waste, the carbon emissions associated with their production can be **lower than those of fossil-based jet fuel**, especially when sustainably sourced and efficiently processed.

As said before, the main reason why aviation today chose SAF is for its **drop-in capacity**, meaning that they can be **blended** with standard jet fuel and used in **existing aircraft engines**, airport **fueling systems**, and fuel **logistics networks** without major modifications as seen in *Figure B* [6]. This makes them especially important in the aviation sector, where replacing the entire fleet or infrastructure would be **extremely costly** and **technically complex**. In this sense, SAFs offer a practical near-term decarbonization pathway, being **introducible progressively** within the existing aviation ecosystem rather than requiring a complete technological transition.

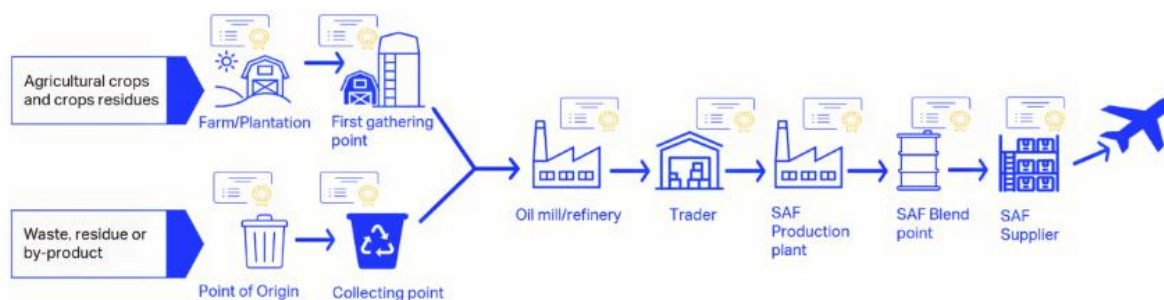


Figure B - Sustainable Aviation Fuel Production and Supply Pathway

For these reasons, SAFs are viewed as one of the **most realistic solutions** available today for lowering emissions from commercial aviation while allowing the industry to continue operating within its **existing technical capabilities**.

C – Current use of SAF

In a December 2023 report, IATA values that year's SAF production at around 600million liters: an impressive growth when compared to 2022's 300million liters, but still a minuscule portion of the total industry at 0.2% of total jet fuel use. [7] [8] Let us now dive into this current 0.2% and better understand how SAF is being implemented.

As stated in section I.B we will focus on **HEFA** biofuels, being the **most common** sustainable fuel alternative today, primarily due to its usage main advantage: drop-in easiness. While some alternative propulsion technologies require heavy technological and infrastructures modification if not entire aircraft design re-conception, HEFA SAFs can directly be implemented in modern turbofan airliners fuel tanks similarly to conventional jet fuels. This **“drop-in” characteristic** represents a high-value advantage, being financially and technically attractive to airlines who started implementing SAF-Fossil fuel blends commercial routes. These implementations come in contracts with major renewable biofuels companies such as SkyNRG or Shell. Test campaigns of such blends began in 2008 in the ground test of a CFM 56-7 B using 50% conventional jet fuel and 50% biofuel partly composed of biomass algae and are **now present in operating commercial routes** across the world. Such test flights were incorporated in various route types, ranging from **short haul to long haul flights**. For example, Air France, Airbus & SkyNRG partnered in October 2011 to fuel A321 operating the Toulouse – Paris route with 50% used cooking oil blends. On the other side of the commercial route's spectrum, Singapore Airlines and others partnered in May 2017 to fuel A350-900 operating the San Francisco – Singapore route with 50% cooking oil blends. [9] From the same document extract, below is shown a summary of biofuel test campaigns internationally.

Observing appendix 2, we can see clearly see how modern day **SAF's implementation** issue an **international subject** for the **global aviation** industry is, with a more intensive focus in Europe, ME and SA regions.

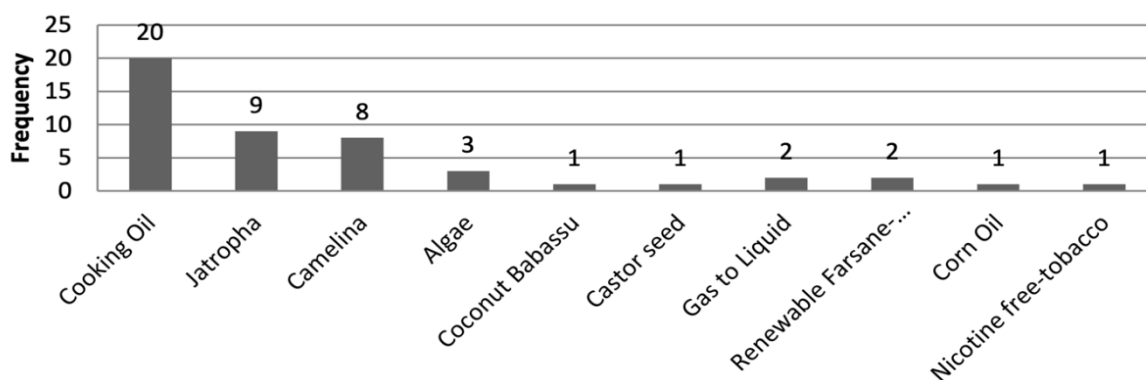


Figure C - Type of Biojet fuel in test flights

Furthermore, we can visualize how the current scope of SAF usage is clearly focused on HEFA primarily, which we recall are produced from vegetable and animal fats / oils (therefore including cooking oils). Still from the same reference, the figure below displays the number of test flights realized per biofuel type (excluding more than one biofuel blends).

To summarize, the current use of SAF blending in modern commercial aviation **primarily focuses on HEFA** type biofuels, which are blended with conventional jet fuels **up to 50%** for the great majority of cases. This blend limit is amongst one of the main challenges of today's SAF implementation and will be presented and analyzed in the rest of this document.

D – Main challenges

Nowadays, SAFs seem to be the **most promising sustainable fuel for aviation**. Despite this large potential, the worldwide use of SAFs at high blend ratio is currently facing some challenges. Those challenges come out on the technical side, the mechanical side and also on the economic side.

SAFs are so chemically close to Jet A-1 that they can easily be blended in the fuel. As their burning power is higher than this of the Jet A-1, it seems beneficial for aviation to blend as much as they can of SAFs in their fuel. But when we look closer, we figure out that we only blend them at 50 %. So where does this limit come from?

In 2023, the **Emirates A380 flight** and the **Virgin Atlantic Boeing 787** have proven the technical feasibility of operating aircraft on **100% SAF**. So, we know that such a blending in the engine is possible. But those were only **experimental cases** and 100% SAFs flights do not exist beside those experiments. Actually, to reach 100% SAF, **modifying your engine becomes mandatory** and this is why we only stay at 50% *“The compatibility of biofuels with jet engines has been extensively tested, demonstrating that biofuels can replace conventional jet fuel up to a 50% blend with minimal to no modifications required to the engines”* [10]. This limit is the sum of unresolved material compatibility and fuel property concerns that we will examine more in detail in our technical study. The most critical aspect is the **lack of aromatics in SAFs**.

Conventional aviation engines were designed to support Jet A-1. This fuel has the characteristic to comport many aromatics. Those aromatics play an essential role in the well-going of the engine since they permit, once the fuel is burned, to **make the joints called O-rings to swell**. This phenomenon makes the engine sealed. Since those aromatics are supposed to make your components totally hermetic, an usual engine at more than 50% blending would be damaged because of the **non-swelling of the components** and the creation of **fuel leaks** in the system. It will be developed further in the technical study.

Another immediate limitation of SAFs use its **density**. As explained in ... *“While piston and turboprop engines may require some adjustments for optimal biofuel performance, turbojet and turbofan engines have shown a higher degree of readiness for biofuel integration,*

particularly with SAFs” [10]. But compared to the Jet A-1, their density is lower. For actual aircrafts, fuel tank capacities are fixed, therefore, the use of SAF at high percentage will directly induce a **reduction in the range or payload capacity**. This is a major challenge that aviation is facing since most frequent flights are mainly medium-haul and long-haul.

One last major challenge is the economic viability of SAFs. Compared to Jet A-1, the SAFs production is **way more expensive**. While a **ton of Jet A-1 costs around 734€**, a **ton of HEFA costs 1 461€** [11]. Noting that the **HEFA production’s method is the cheapest**, this explains why HEFAs are mainly used in aviation. Also to produce those HEFAs, you need a large amount of land dedicated to production of farming wastes which is for now not possible to have. *“Feedstock prices, especially for bio-based SAF derived from feedstocks like used cooking oil or algae, contribute to production costs”* [12].

II – Technical Study and Analysis

After establishing a relevant state of the art and formulating our problem clearly, this part will focus on a technical analysis of the problem to bring a qualitative response to whether or not, SAFs and especially HEFA-SAF can be blended at more than 50% in modern engines. We will discuss the **benefits** and **drawbacks** of those incorporations to help us take a decision.

This section will include a **regulatory part** to check the limit of SAFs application to aviation field regarding the existing norms and rules. Then, we will have a **thrust analysis depending on the dilution ratio of SAF with JetA-1** to explore the impact on performances. Another analysis on the **structural impact of SAFs** in modern airplanes to check the needed volume to carry SAFs. Finally, we will see a **practical case** on the current utilization of the 50% SAFs blending nowadays.

A – Regulatory compliance of SAF

1. Regulatory Framework and JIG Roles

Before conducting a technical analysis of Sustainable Aviation Fuels, it is essential to define the regulatory framework that governs their use. For this study, we rely on the **JIG Bulletin 141** (April 2022), which provides the **AFQRJOS Checklist** (Issue 33) [13]. This document is the global benchmark for jet fuel quality; it acts as a **synthesis that harmonizes the requirements set by two organizations**; the American **ASTM with D1655** and the British Ministry of Defense via **DEF STAN 91-091**. The American Society for Testing and Materials (ASTM) is the international leader in developing technical standards for a wide range of materials. The British Ministry of Defence issues the DEF STAN to ensure rigorous fuel specifications for both military and civil aviation.

Starting with the analysis of this regulation is mandatory as in aviation, a fuel's viability is not only measured by its performance but firstly by its **compliance with airworthiness standards**. The JIG Checklist can be seen as a safety filter; it can determine whether a SAF can be certified as a 'drop-in' fuel. As a result, this document will be **our validation reference**. Each technical parameter such as density or heat of combustion must be within the strict range defined by the checklist to **ensure operability and safety**.

2. Analysis of JIG Callout specifications Document and Key Regulatory Limits for SAF

By analyzing the JIG Checklist, we can identify the **technical boundaries** that define the current 50% blending limit. Even though SAF is designed to emulate conventional kerosene, several critical parameters still prevent its use at 100% concentration under current regulations.

i. Blending limits: fuel density, aromatics content and freezing point

The first major issue is **fuel density**. According to the JIG standards, any fuel loaded into an aircraft, including SAF blends, **must be certified as Jet A-1**, which requires a final density **between 775.0 and 840.0 kg/m³**. Since many pure SAF components presents densities **as low as 750 kg/m³**, they fail this requirement on their own. Consequently, blending them with conventional kerosene is necessary to bring the final mixture back into the regulatory range.

Furthermore, while the JIG Checklist mandates a maximum aromatic content of 25.0%, it explicitly requires compliance with **DEF STAN 91-091** [14] for the fuel to meet aircraft operating limitations. As stated in Note 114 of the Bulletin 141:

‘Jet A-1 fuel will only meet the aviation fuel operating limitations of airplanes certificated to operate on Jet A-1 fuel and the requirements of Def Stan 91-091 if [...] the fuel has been tested for aromatics concentration’.

Although the JIG table only displays the upper limit, the reference DEF STAN 91-091 establishes a **minimum range of 8.0% v/v**. This is a critical safety requirement because aromatics ensure the **swelling of nitrile rubber seals** in the fuel system. Without this swelling, the risk of fuel leaks increases. Since most SAF production methods result in fuel with **near-zero aromatics**, the chemical composition is the primary reason why the JIG maintains a strict 50% ratio on synthetic components.

To visualize these constraints, a numerical model was developed to simulate the **evolution of fuel properties based on the blending ratio** (from 0% SAF to 100%). To do so, we use the linear blending rule, assuming that the physical and chemical characteristics are proportional to the volumetric ratio of each component. With x the concentration of SAF, ρ_{mix} the mixture density and A_{mix} the aromatics content:

$$\rho_{mix} = x \times \rho_{SAF} + (1 - x) \times \rho_{JetA1}$$

$$A_{mix} = x \times A_{SAF} + (1 - x) \times A_{JetA1}$$

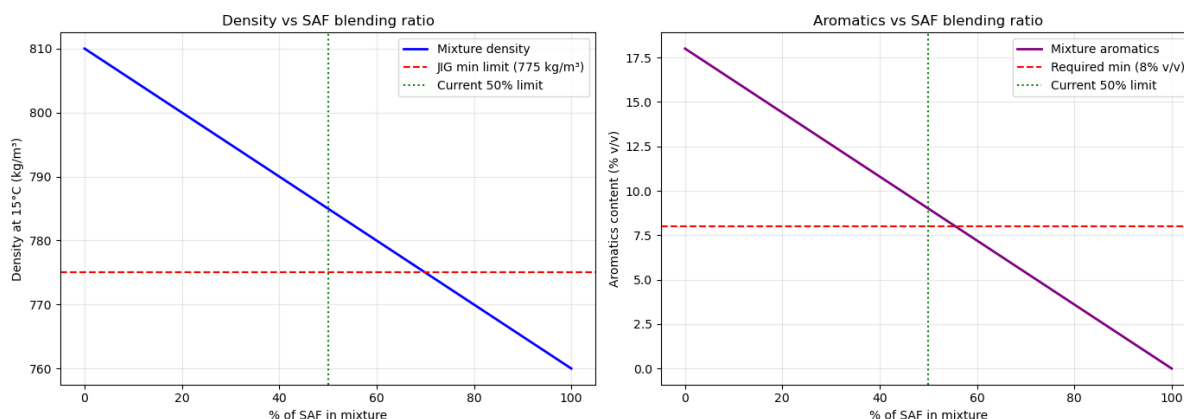


Figure D - density and aromatics depending on the blending ratio

As shown in the figure D, increasing SAF concentration pushes density and aromatics toward critical regulatory limits. For this simulation, we used a ρ_{JetA1} of **810 kg/m³** and a ρ_{SAF} of

760 kg/m³, a typical value for **HEFA-SPK** according to **ASTM D7566** standards referenced in the JIG Checklist. At 100% SAF, the density fails the 775.0 kg/m³ minimum requirement. Moreover, with a standard Jet A-1 at 18% aromatics, a 50% blend results in a **9% content** which is barely above the 8% limit. This simulation confirms why the 50% limit is a **necessary safeguard for current airworthiness**.

Beyond chemical composition, the freezing point is a fundamental safety requirement under JIG Bulletin 141, mandating a max. of **-47.0°C** for Jet A-1 certification to prevent fuel waxing at high altitudes. However, as demonstrated in the study led by L. Starck and al. [15], achieving this performance involves a trade-off. To maximize SAF yields and reduce production costs, manufacturers might produce 'relaxed' batch of higher freezing point (-27°C).

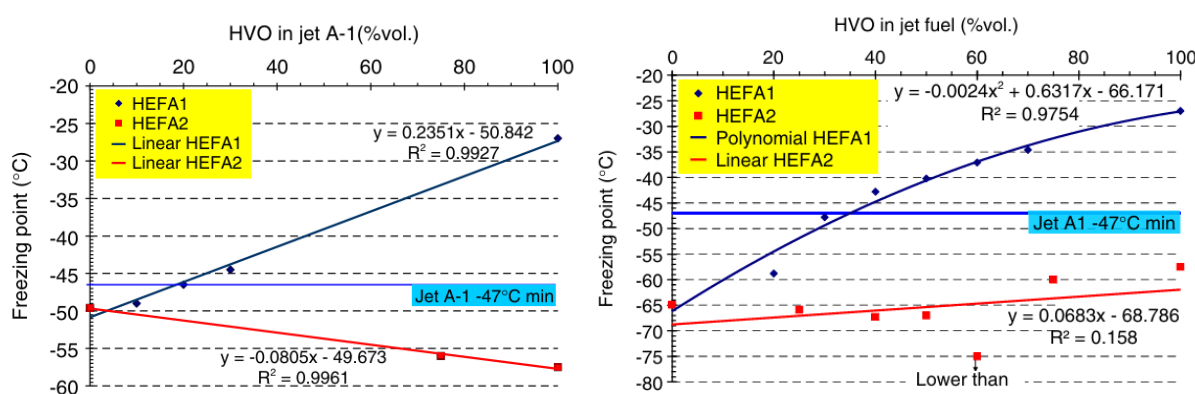


Figure E - impact of HEFA content on freezing point for blends with conventional (left) and high-quality (right) Jet A-1

Figure E highlights that the maximum blending ratio depends heavily on the base kerosene quality. In Figure E, using a standard Jet A-1 (-49.6°C) mixed with 'relaxed' HEFA, the -47°C limit is crossed at only 16% SAF concentration. Conversely, it shows that using high-quality Jet A-1 (-64.9°C) allows the same 'relaxed' SAF to be blended up to 35%.

However, it is important to note that reaching the 50% regulatory limit, or even exceeding it theoretically is possible when using high-quality, fully isomerized SAF, such as HEFA-SPK 2 (-57.5°C). Unlike 'relaxed' versions, this high-quality SAF does not affect the cold-flow properties of the blend, proving that the 50% ceiling can be reached through a trade-off between production and quality control.

ii. Advantages: Low Heating Value and thermal stability

On the other hand, SAF demonstrates superior qualities in other areas. For instance, the JIG requirement for Low Heating Value (LHV) is a minimum of **42.8 MJ/kg**. As our study will confirm, the SAF higher energy content with an LHV of around **44 MJ/kg** directly influences engine performance like **the Specific Fuel Consumption (SFC)**.

In addition to that, SAF demonstrates **excellent thermal stability**. It refers to the fuel's ability to resist chemical decomposition when exposed to high temperatures within the engine's fuel system. While conventional kerosene can create carbon deposits that clog fuel injectors, SAF exhibits higher thermal stability, as verified by the JFTOT test in the checklist. This is due to its high purity and the absence of sulfur or other possible contaminants, which leads to **cleaner combustion** and **less engine maintenance**.

B – Thermodynamic study: dilution impacts of SAF on the aircraft's thrust performances

1. Scope of the study

As established in our state of the art, **airworthiness regulations** (specifically ASTM D7566) currently cap the SAF blending ratio at **50%** by volume for commercial operations. This limitation is primarily driven by **material compatibility** concerns, most notably the effect of reduced aromatic content on the **swelling** of fuel system seals. However, for the purpose of this thermodynamic and physical analysis, we will lift this technical constraint.

We assume here that the blending ratio can vary freely from 0% (pure Jet-A1) to 100% (pure SAF). This assumption allows us to observe thermodynamic trends across the full spectrum, even though this would technically require **minor adjustments** to engine sealing.

2. Characteristics of HEFA-SAF

A key characteristic of HEFA-SAF is its Lower Heating Value, that we will call LHV, which is approximately 2.3% higher than that of **fossil kerosene**. For HEFA we reach 44.2 MJ/kg where Jet A-1 reaches 43.1MJ/kg. On paper, this means that for the same mass, HEFA-SAF releases more energy during combustion. We might instinctively conclude that the **overall energy** released from combustion will be **better** and that **SAF is preferable** for increasing **thrust**. But is this intuition always verified under real operating conditions?

3. Definition of the studies

We will verify this by comparing two distinct regulation modes. Two complementary approaches were used for this study:

*A- A thermodynamic model developed in **Python** to isolate physical variables :*

A one-dimensional cycle model of a dual-spool unmixed-flow turbofan was implemented in Python, based on the Brayton cycle framework. The model computes the thermodynamic state at each station of the engine and derives the net thrust and specific fuel consumption (SFC) for a given set of operating conditions.

*B- A thermodynamic model on the professional software **GasTurb 15** :*

GasTurb is a professional gas turbine performance simulation software widely used in academia and industry. It provides a higher-fidelity cycle model with component maps, cooling flows, and bleed air systems. In this study, GasTurb was used in Design Point Parametric mode to sweep the fuel Heating Value from 43.124 MJ/kg (Jet-A1) to 44.1 MJ/kg (100% HEFA-SAF), with all other parameters held constant.

4. Assumptions

All simulations are conducted at cruise conditions and the Python model relies on the air standard approximation (constant c_p and γ), allowing us to focus on the relative variations induced by the fuel rather than absolute certification values. Therefore, we have the next table:

Property	Jet-A1	HEFA-SAF
LHV (Lower Heating Value)	43.1 MJ/kg	44.2 MJ/kg
Density (at 15 °C)	~800 kg/m ³	~760 kg/m ³

Table 1 – Properties values

Parameter	Value
Altitude	11 000 m (ISA standard)
Static pressure	0.47×10^5 Pa
Static temperature	249 K (-24 °C)
Flight Mach number	$M \approx 0.8$ ($u = 250$ m/s)

Table 2 – Parameters values

5. Python simulation

We then plot our results using our Python simulation of the turbofan engine. We choose to plot the Thrust, as well as the specific consumption, in function the dilution ratio x . For that, we mix our LHV values depending on the x parameter. Since the ratio is a volume one, we use this equation to get the x_{mass} ratio to later mix the LHV:

$$x_{mass} = \frac{x_{vol} \cdot \rho_{HEFA}}{x_{vol} \cdot \rho_{HEFA} + (1 - x_{vol}) \cdot \rho_{JetA1}}$$

We obtain the following figure:

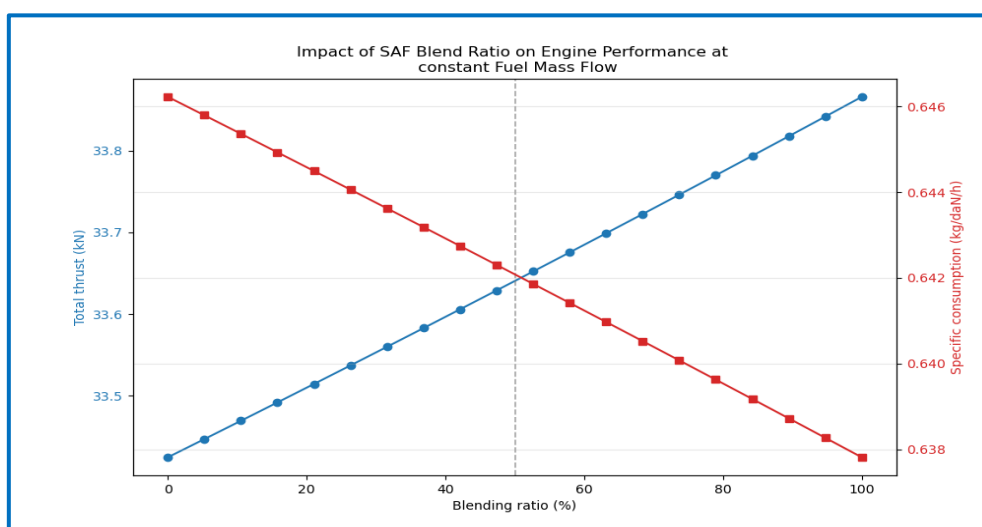


Figure F – Impact of SAF Blend Ratio on Engine Performance at constant Fuel Mass Flow.

In this first phase, conducted via Python, we simulate a scenario where the pilot maintains a fixed throttle position: the engine receives exactly the same mass of fuel per second (0.6 kg/s), regardless of the blend. Since the LHV is higher, the temperature at the combustion chamber exit rises. We observe a nearly linear increase in net thrust : **+1.32% for 100%HEFA-SAF** and a symmetric drop in Specific Fuel Consumption (SFC).

If we only looked at the mass flow, we would use 100% SAF on every airplane in service to gain thrust. However, this simulation is neglecting a major constraint which is the material integrity.

In reality, engine manufacturers impose a maximum Turbine Entry Temperature (TET) usually between 1450 K and 1500 K, to prevent melting or accelerated fatigue of the turbine blades.

6. GasTurb15 simulation

This is why we conducted a second simulation on GasTurb15. By using GasTurb15 with a fixed TET condition (1450 K), the results are reversed. Because HEFA-SAF has a higher LHV, the engine reaches the temperature limit with a lower mass flow rate in fuel. Therefore, the net thrust resulting from this numerical adjustments becomes lower as we increase the blending ratio. Here is the results we get from the GasTurb15 simulation :

Since we could only access a demo-version of GasTurb15 for 14 days, the file cannot be included in the final folder.

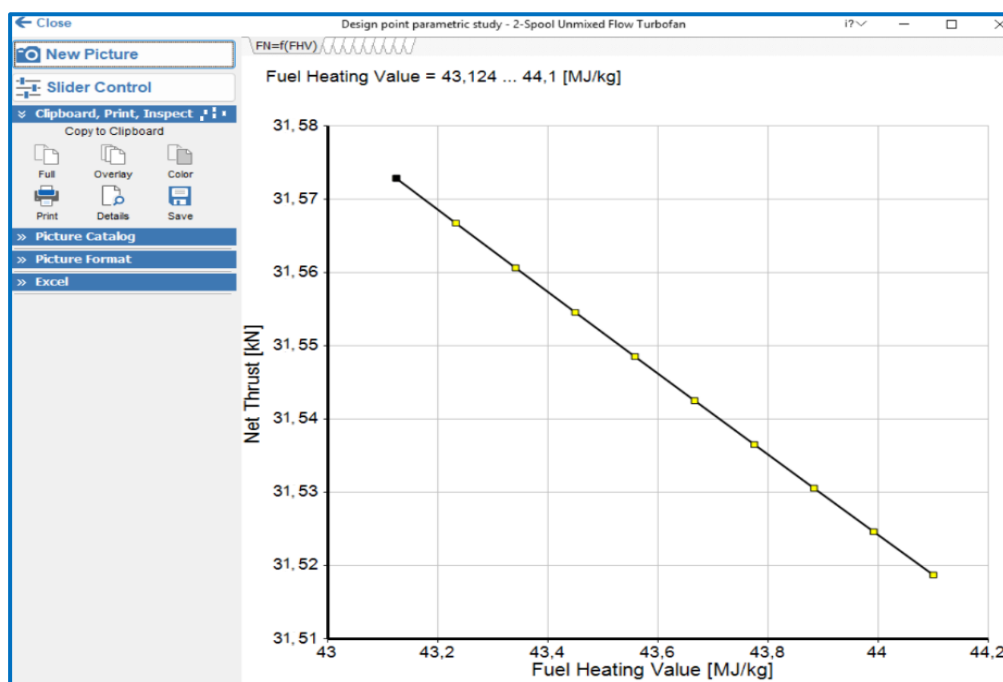


Figure G – Impact of SAF Blend Ratio on Engine Performance at constant TET.

7. Results interpretation

The conclusion of this study shows that the impact of SAF on performance depends entirely on how the engine is regulated. We can conclude that in reality, we will stick to the second scenario and not choose to damage the turbine only for a 1.3% rise in thrust. Instead, we will fix the TET and have a small loss in thrust but the gain in consumption is beneficial.

From an operational standpoint, these variations remain below 1.5%, explaining why SAF is considered a "Drop-in" fuel. It is practically transparent for aircraft performance in cruise, requiring no changes to flight procedures to maintain the required thrust.

C – Thermodynamic study: dilution impacts of SAF on the aircraft's range performances

1. Scope of the study

After concluding that the initial attractivity of HEFA-like SAFs operation is quickly balanced by its TET related challenges for engines lifespan, let us tackle another negative impact of 100% SAF usage theoretical implementation : the aircraft's range for a fixed fuel capacity.

Indeed, recalling back the table 1 values, we observe that the lower density of HEFA penalizes more this fuel type despite its higher specific heat value. Knowing that aircrafts fuel quantity is maxed by fuel tanks volumetric capacities, we can transform each fuel energetic potential into volumetric specific capacities.

Note : for this study, more precise density values will be used from a CERFACS large-eddy simulation report - $\rho_{HEFA} = 0.752 \text{ kg/L}$ & $\rho_{JETA1} = 0.793 \text{ kg/L}$. [16].

Property	Jet-A1	HEFA-SAF
Volume Specific LHV (at 15 °C)	34.2 MJ/L	33.2 MJ/L

Table 3 - Properties refined values

Observing these values lead us to a relevant concern : does this **energy-per-volume discrepancy** of HEFA-like SAFs **translates to an operational limit** for 100% HEFA usage ?

2. Definition of the study

To assess this question, we will calculate and **compare** a chosen aircraft **configuration's range** flying on **100% Jet-A** conventional fuel and **100% HEFA** alternative fuel.

To create this configuration, we will pick one of the most popular modern medium-haul airliners : the Airbus A320Neo lifting off at MTOW. The rest of the parameters will be either imposed by the manufacturer datasheet's specificities or chosen from different configuration possibilities of these specificities. Each of these choices were made to represent the most generic configuration possible. [17] [18] [19].

Parameter	Ref - Value	Data source type
Aircraft type	A320neo	Chosen
Weight configuration	WV050	Chosen from manufacturer spec.
Max Take Off Weight (MTOW)	73,5 t	Manufacturer Spec.
Operating Empty Weight (OEW)	44.3 t	Manufacturer Spec.
Max Zero Fuel Weight (MZFW)	62.8 t	Manufacturer Spec.
Engine	CFM LEAP 1-A	Chosen from manufacturer spec.

Parameter	Ref - Value	Data source type
Cruise Mach (M)	0.78	Manufacturer Spec.
Cruise Altitude (z)	FL365	Chosen from manufacturer Spec.
Lift/Drag ratio (f)	15.67	Manufacturer Spec.
Total mass per passenger	95 kg	Chosen
Engine Thrust Specific Fuel Consumption for Jet-A-1 engine operation (TSFC)	14.4g/kN/s	Manufacturer Spec.

Table 4 – Second study parameters value.

3. Assumptions

The main assumption used for this study is the Breguet integrated range equation:

$$R = \frac{2 \cdot v \cdot f}{TSFC \cdot g} (1 - \sqrt{1 - Q}) \quad \text{with} \quad Q = \frac{m_{fuel}}{m_{initial}} = \frac{m_f}{MTOW}$$

This expression, approximated from the real integration expression for range (as mass changes with fuel consumption) and not considering Take-Off and landing aircraft configurations, represents the major assumption of our study. It deviates from continuous integration range results by around 10% for our typical resulting range value, deviation which will be considered to validate or not our results compared to Airbus's A320Neo range charts. [20]

However, we could also note our flight altitude FL365 (in reality increasing as the aircraft gets lighter) or mass per passenger, both of which are arbitrarily chosen averaged from typical practical values as they do not represent a major impact on the study's result relevance.

4. Range calculation & Comparison

To first tackle our calculation, let us isolate the fuel mass of our aircraft configuration.

$$MTOW = OEW + m_{payload} + m_f = MZFW + m_f \Leftrightarrow m_f = 73.5 - 62.8 = 10.7 \text{ t}$$

As stated previously, even though this exact configuration is maxed by the manufacturer MTOW masses for simplicity, we know that aircraft weight capacities are limited by volume. Considering that, we can calculate the equivalent mass of HEFA fuel our A320Neo could embark at the same volume that this 10.7 t represents for conventional fuel.

$$V = \frac{m}{\rho} = \frac{m_{f-A1}}{\rho_{A1}} = \frac{10.7E3}{0.793} = 13493L \Rightarrow m_{f-HEFA} = \rho_{HEFA} \cdot V = 0.752 \cdot 13493 = 10.15 \text{ t}$$

The speed variable is directly derived from the altitude temperature at FL365 (ISA conditions).

$$v = M \cdot c = M \cdot \sqrt{\gamma r T} = 0.78 \sqrt{1.4 \cdot 287 \cdot (288,15 - 0.0065 \cdot 36500 \cdot 0.3048)} \approx 229,6 \text{ m/s}$$

For the value choice for TSFC a Jet-A-1 fuel and a HEFA operating engine, our approach will be as follow : stick with the standard CFM LEAP 1-A TSFC known value of 14.4g/kN/s (operating on conventional fuel) and estimate a ratio between its HEFA-operating counterparts. To do so, let's recall some flight mechanics and thermodynamics power equations:

$$P \propto T \cdot v \quad \Rightarrow T \propto \frac{P}{v} \propto \frac{\dot{m}_f \cdot LHV}{v}$$

$$P \propto \dot{m}_f \cdot LHV$$

Furthermore:

$$TSFC = \frac{\dot{m}_f}{T}$$

Giving us:

$$TSFC \propto \frac{v}{LHV}$$

And with the configuration speed being identical in both fuel usage, this allows us to establish:

$$TSFC_{A1} LHV_{A1} = TSFC_{HEFA} LHV_{HEFA}$$

$$\Leftrightarrow TSFC_{HEFA} = TSFC_{A1} \frac{43.1}{44.2} = 0.975 \cdot TSFC_{A1}$$

Comparing this new property difference between Jet-A-1 and HEFA usage, we can see the emergence of this **study's core trade-off analysis** :

➤ **SAF's better fuel efficiency VS worst density.**

We therefore have, for our given configuration :

$$\diamond \frac{2 \cdot v \cdot f}{TSFC \cdot g} (1 - \sqrt{1 - Q}) = \frac{2 \cdot 229.6 \cdot 15.67}{14.4 \cdot 10^{-6} \cdot 9.81} (1 - \sqrt{1 - \frac{10.7}{73.5}}) \Leftrightarrow \boxed{R_{A1} = 3854km}$$

$$\diamond \frac{2 \cdot v \cdot f}{TSFC \cdot g} (1 - \sqrt{1 - Q}) = \frac{2 \cdot 229.6 \cdot 15.67}{0.975 \cdot 14.4 \cdot 10^{-6} \cdot 9.81} (1 - \sqrt{1 - \frac{10.15}{73.5}}) \Leftrightarrow \boxed{R_{HEFA} = 3741km}$$

Before jumping to conclusions, we can validate the coherence of our model by correlating our Jet A-1 range value with Airbus's A320Neo data sheet (appendix 4). Considering the Breguet range deviation of 10% at our range value of 3800km~2000nm (appendix 3), we confirm that our R_{A1} for our configuration indeed correlates with the manufacturer's value.

Once that done, we can observe that the **100% HEFA configuration**, with lower TSFC but **lower density**, fell for its latter parameter with a **113km shorter range** than the JetA1 flight.

Lastly, we could modify our study with a V_{fuel} fixed at the A320Neo's **maximum** of 26700L. By varying the payload similarly to Airbus's charts and for the same MTOW limit, we obtain the following :

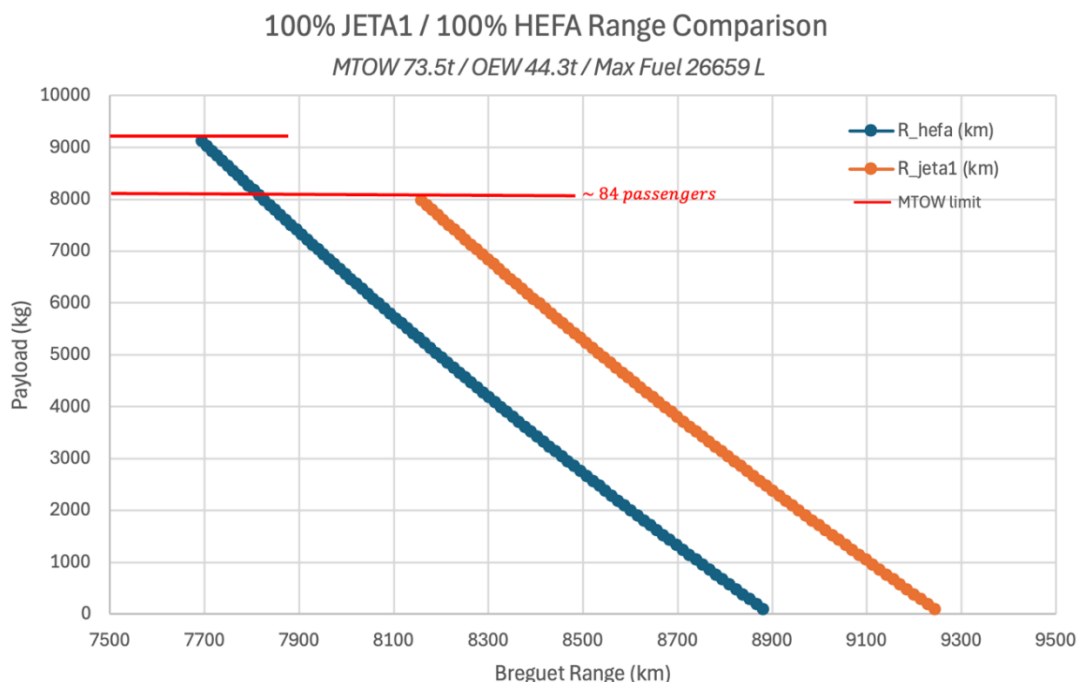


Figure H – Second study parameters value

Despite the fact that such configuration does not represent the typical operational use of the aircraft (more oriented towards medium range with most passengers possible), we nonetheless observe the **far greater range differences at lower payloads**, +350km short for HEFA.

5. Results interpretation

To summarize, this new technical approach of our subject presented us with another operational constraint implied by the theoretical implementation of SAFs as conventional jet replacement in current airliners: **structural fuel capacity ceiling**.

This challenge represents a core problematic:

- **Addressing it**, by matching the SAF fuel mass capacity of an airliner to its JET-A conventional configuration, demands a non-negligible fuel tank volume increase. In the extremely optimized machines that modern commercial aircrafts are, imposing such a demand represents a fundamental structural design re-evaluation : **a high financial and temporal costing** challenge. Contrary to recent decarbonization airliners modifications such as winglets or engine replacement, increasing internal fuel tanks volume requires to review **nearly from scratch the conception of the airplane**.
- **Not addressing it**, yet still aiming considering 100% SAF usage, will negatively impact on the operational range possibilities. In the high-cost low margin business that commercial air transportation is, such a constraint represents a **very imposing if not impossible change** for future profitability of this market's actors.

D – Study of current applications with a mix over 50%: B787

1. Case of the study

While the current regulatory framework maintains a strict 50% blending limit for commercial aviation, recent experimental milestones have sought to prove the **feasibility of 100% Sustainable Aviation Fuel (SAF)** operations. A notable application in this field is the Virgin Atlantic "**Flight100**" performed in November 2023. This mission involved a Boeing 787-9 Dreamliner equipped with Rolls-Royce Trent 1000 engines, flying from London Heathrow to New York JFK using a **total "drop-in"** fuel load with no airframe or engine modifications. [21]

2. Technical implementation

The technical success of this application relies on the management of specific thermodynamic properties identified in our study. As established in the thermodynamic simulations, **HEFA-type SAF** possesses a **higher** Lower Heating Value (LHV) of approximately 44.2 MJ/kg **compared to** the 43.1 MJ/kg of conventional **Jet A-1**. In a scenario where the fuel mass flow remains constant, this **higher LHV leads** to a linear **increase** in net **thrust** of approximately **1.32%** and a corresponding **drop in** Specific Fuel Consumption (SFC). However, **to preserve material integrity**, engine manufacturers impose a **maximum** Turbine Entry Temperature (TET), typically **between 1450 K and 1500 K**, to **prevent the melting or accelerating fatigue** of turbine blades. Consequently, when operating at a fixed TET, the engine reaches its thermal limit with a **lower mass flow rate of SAF**, resulting in a **slight loss of net thrust** but a **beneficial gain in consumption**.

The most **critical challenge** for 100% SAF implementation is the **absence of aromatics**, which are **essential** for ensuring the **swelling** of nitrile rubber **O-ring seals** in the fuel system. Without this swelling, the engine is not properly sealed, creating a **risk of fuel leaks**. This chemical deficit was addressed in the B787 mission by using a **specific blend** that included Synthetic Aromatic Kerosene (SAK), **providing** the necessary **chemical properties** to **maintain** hermeticity without fossil kerosene. [22]

3. Commercial applicability and barriers

Despite the technical feasibility demonstrated by the B787, **several barriers** prevent immediate commercial applicability. From an operational standpoint, the **lower density of SAF** (approximately **760 kg/m³** versus **800 kg/m³**) imposes a structural **fuel capacity ceiling**. Because aircraft fuel tanks have fixed volumetric capacities, using **100% SAF** results in a **lower total mass of fuel** onboard, leading to an operational **range reduction**. Furthermore, the economic "Green Premium" remains a **major deterrent**; SAF production is **currently 2 to 5 times** more **expensive** than kerosene. This significant **price gap** threatens airline profit margins and likely necessitates a structural **increase** in ticket prices. Ultimately, the B787 application confirms that while the industry is technically prepared for high SAF ratios, the transition remains a **financial and structural struggle** rather than a purely technological one.

III – Non-Technical Overview

The previous technical analysis confirms that 50% SAF is not only in compliance with JIG Bulletin safety standards but also provides an important advantage in engine performance with a reduction in specific fuel consumption.

In the aviation industry, a project's viability is dictated by **various non-technical constraints**. To fully address our study of the **multi-domain impact of SAF**, it is essential to look beyond the technical aspects and consider the real-world barriers. Therefore, we will analyze the SAF integration with three different perspectives. First, we will evaluate the **global environmental impact** of SAF. Then, we will analyze the **economic reality of production costs**. Finally, we will examine the **geopolitical stakes of energy sovereignty**.

A – Environmental impact: a way to limit toxic emissions

The main purpose of SAF's adoption in the aviation field is its capability to **decouple aviation growth from an exponential CO₂ emissions**. We know that the **Jet A-1**, once **burned**, introduces a **un-negligeable amount of carbon dioxide** in the atmosphere while already emitting a lot for its extraction and refining. But by using SAFs such as HEFA, we try to rely on a **carbon circular logic**. This production method uses feedstocks like **wasted oils** and **agricultural residues**, this way of working **only recycles** already existing carbon in the biosphere.

If we look closely into the **Life Cycle Assessment (LAC)** of the HEFA, we understand that the impact is **massive**. From the **production** to the emissions due to **combustion**, the carbon dioxide **emissions are drastically reduced**. Current data from bodies like IATA and ICAO shows that SAF can cut net CO₂ emissions by **up to 80%**. This percentage is extremely enthusiastic for the reaching of the **Net Zero emissions goal by 2050**, especially since this type of fuel works well with existing engines.

It is also important to not only talk about the CO₂ emissions benefits of this type of fuel. Since the **chemical composition** of HEFA contains almost **no aromatics and sulfur**, their **combustion** is way cleaner. This contributes to the **limitations of soot particles drop** in the atmosphere. We know that those particles contribute to formation of **contrails**, which contribute to **radiative-forcing**. By decreasing those contrails, we also **limit the emission of SO_x and NO_x**.

B – Economic impact: an expensive alternative

While SAF is technically viable, its integration into the aviation market faces a major economic difficulty: the significant price gap compared to conventional Jet A-1. Currently, SAF is estimated to be **2 to 5 times more expensive than fossil kerosene**, depending on the production method. Since fuel typically accounts for **25% to 35% of an airline's total operating costs**, any increase in fuel price directly threatens the industry's profit margin and overall financial stability.

If there were a massive shift toward high SAF blending ratios, it would inevitably lead to a surge in airfares. For example, a 50% blend, despite its performance benefits, would increase the cost per passenger-kilometer, which could reduce global demand for air travel. This phenomenon, known as “Green Premium” remains the main deterrent for airlines operating in this highly competitive market where price remains a **key differentiator** for customers.

To overcome this, in Europe, the **ReFuelEU Aviation initiative** was implemented. By introducing mandatory SAF blending quotas, the goal is to ensure a stable market for producers, giving them the confidence to build larger factories and lower costs with mass production. Furthermore, the **increasing cost of carbon permits** (EU ETS) is designed to make fossil fuels more expensive, slowly diminishing the price gap between kerosene and SAF.

Nevertheless, until SAF is produced on a global scale, the high costs will remain a heavy burden. This financial pressure creates a two-fold impact: airlines must absorb higher operational expenses, while passengers will likely see a structural increase in ticket prices. Eventually, this demonstrates that while the aviation industry has solved technical challenges of SAF, energy transition remains a **financial struggle rather than a purely technological one**.

C – Geopolitical impact: the new strategic resource

Beyond their environmental role, SAFs also raise strategic questions related to **energy security**, **industrial sovereignty**, and the global competition over feedstocks. As a result, SAFs are not only a **technical solution**, but a **geopolitical tool** that could **redistribute the balance** of fuel-importing and fuel-exporting countries. Their development must also be carefully managed to avoid creating new tensions.

Unlike **petroleum**, which creates **strong dependence** on a limited number of exporting countries, **SAF** can be **produced locally** or regionally from available biomass and waste resources. This gives SAF a newly important **strategic advantage**, since it may **reduce** external energy **dependence** and **improve the resilience** of nation fuel supply chains. This issue can be well illustrated in instable times, such as the current tensions and conflicts affecting oil-producing regions like Iran.

The situation in Iran and the broader tensions in the Strait of Hormuz highlight the **strategic vulnerability** of global **oil dependence**. Because a very large share of the world's petroleum trade passes there, any disruption in the area affects international supply chains and triggers **massive shortages** in countries that **rely heavily** on imported oil from there. Countries that are affected first are now facing economic crises with their country's working reserves that are yet

to end. The development of SAF here could improve the **resilience** of these countries by **diversifying** their **fuel sources** and increasing resilience to **geopolitical tensions**. Thus, making the aviation industry **less sensitive** to disruptions such as those currently affecting the Middle East.

However, this strategic potential also comes with an important **limitation**. For **first-generation SAFs**, the use of agricultural land and food-based feedstocks raises a **food versus fuel** issue. If crops are diverted from food production to fuel production, this can increase pressure on already tense food systems, thus contributing to **higher prices, reduced food availability** or **security concerns**. These impacts could greatly **affect** our **everyday lives**, especially in countries where agricultural resources are limited or where food imports are already vulnerable. The impact of SAFs **varies significantly** from one country to another, depending on the **type of feedstock** used, the **scale** of production, and the **local balance** between **energy** needs and **food** needs. Whether SAF production relies on waste, residues, or dedicated energy crops.

In other words, SAF can **reduce dependence** on fossil-fuel-exporting countries, but it may also **create new strategic tensions** if it competes with food production. For those reasons, the development of advanced SAF pathways based on non-food feedstocks is **essential** to ensure that the **transition** to more sustainable aviation **does not generate** new geopolitical or social **risks**.

Executive Conclusion : Client Response

After investigating the multifactorial aspects defining today's feasibility and consequences of SAF usage in commercial aircrafts, we propose the following executive summary of our core problematic. In response to the presented problematic, **we recommend complying with the common 50% blending ratio limit for long-term commercial air transport operations, as it represents a more – economically safe, technically and geopolitically feasible, as well as legally in-lined – operational solution.**

We firstly justify this standpoint by summarizing the technical operability of SAF usage in commercial aircrafts. Having higher heat capacity than conventional jet fuel, the blending or total replacement of the latter by SAF fuels may appear as a strategic advantage as demonstrated in our engine performance study: higher thrust, lower specific consumption... However, these points quickly fall short of the disadvantages associated with this usage. Firstly, combustion heat is limited by the engine turbine blades maximum temperature resistance, already optimized for conventional fuels. Furthermore, operating a new aviation fuel requires certifiability, which represents a hard limit in our case. Indeed, SAFs contain nearly zero aromatics, chemical components necessary to prevent leaks. Furthermore, high percentage (>50%) SAF-Fossil fuel blendings fail many international JIG fuel property requirements such as density, or freezing point limits. It is essential to understand that these requirements do not only represent legal limits but critical safety regulations, where failure to compliance can lead to catastrophic accidents such as blocking fuel transit and compromising engine operability. Lastly, we projected the impact of long-haul flight feasibility operating theoretical 100% SAF usage and demonstrated another challenge: higher aircraft fuel capacity requirements. While such values may appear negligible (hundreds of km of range difference), it is important to recall that commercial aviation represents one of the most high-cost low-margin businesses, and that even such small numbers can pose serious long-term financial and operational impacts.

Secondly, the economical, geopolitical overview of the subject put light on the complex and unstable future of high volume SAF acquirement despite its ecological advantages. Being 2 to 5 times more expensive as conventional fuel, this budget change imposes a major glass ceiling for airlines of which one third of the current budgets represent fuel expenses. While the market is actively being regulated in the hope of stability for decreasing prices, its current state still represents a very imposing setback for high usage commitments. On the other hand, SAF usage and therefore production increase could represent a valuable geopolitical sovereignty advantage for committing nations, cutting down dependencies on fuel-production unstable regions. However, it is important to implement a promotion for non-feedstocks SAF production, limiting the potential food-market competition setbacks. Lastly, it is important to keep in mind that despite all these challenges, SAF today represents the most promising aviation decarbonation pathway technology, estimated by IATA as 65% contribution in achieving Net-Zero aviation by 2050. Therefore, it is essential to consider the previously mentioned setbacks as imperative daunting yet surmountable challenges for significant aviation emission reductions.

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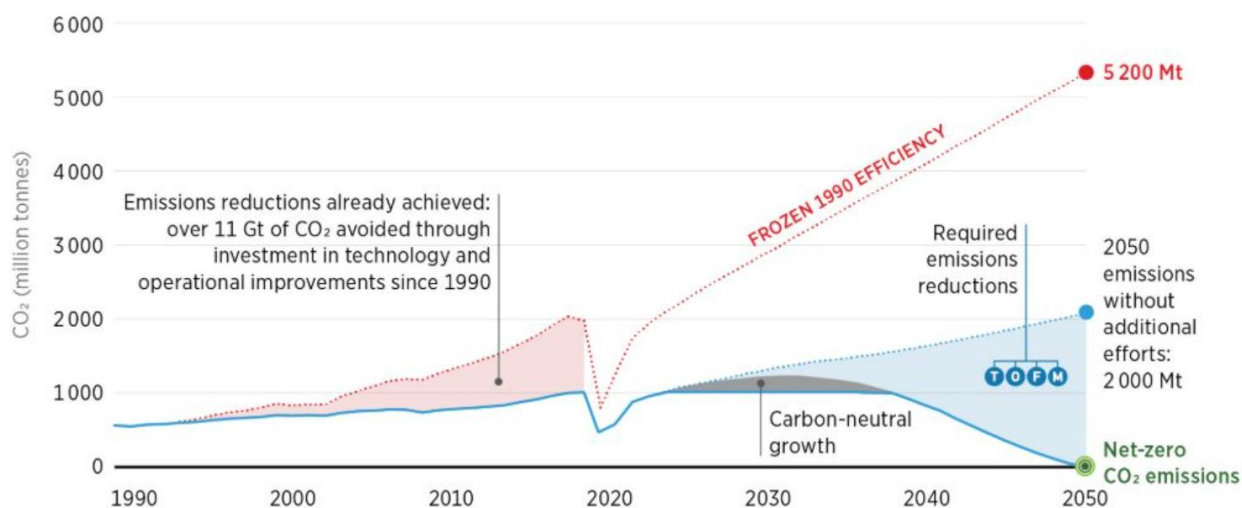
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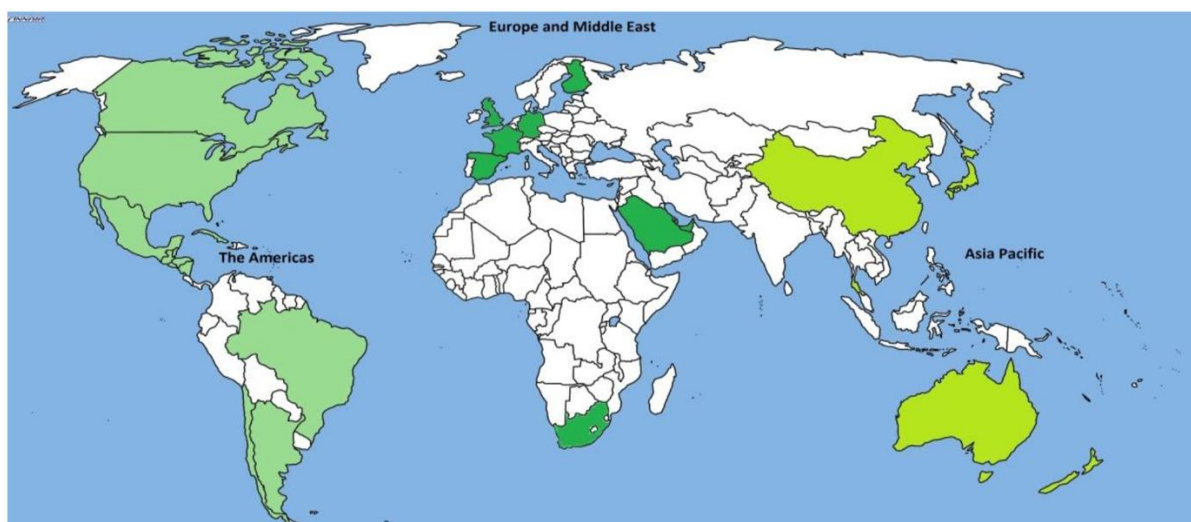
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Appendices

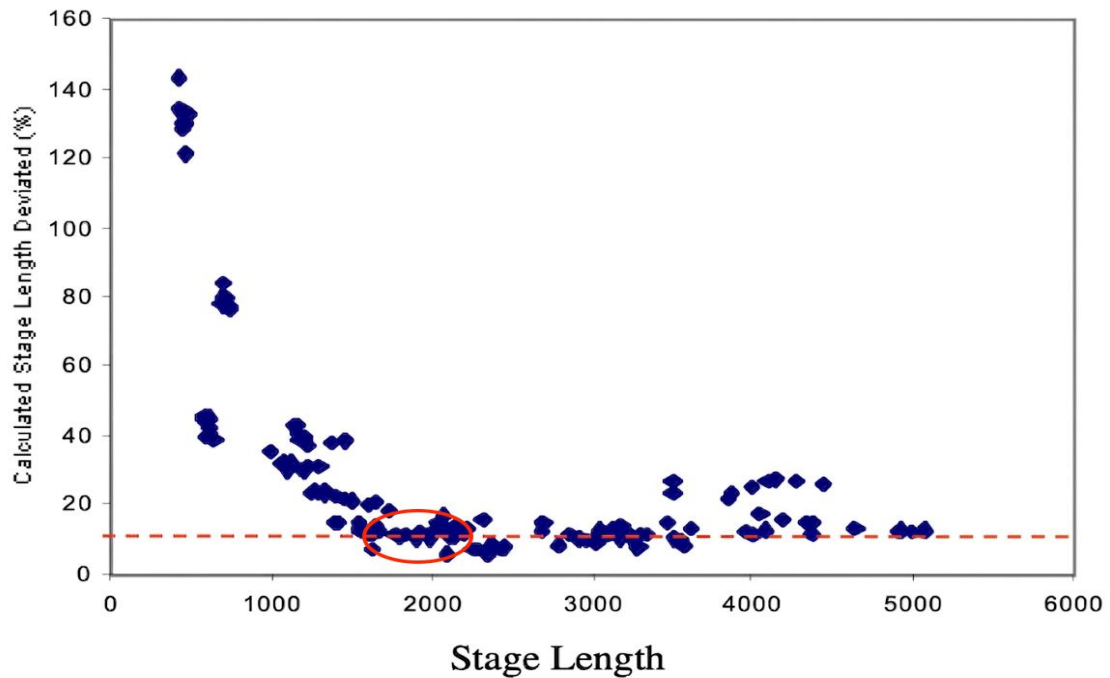
➤ Appendix 1 - Historical & projected CO₂ emissions of the aviation sector, 1990-2050



➤ Appendix 2 – The world biojet fuel map for Commercial Aviation flight



➤ Appendix 3 – Breguet range calculation deviation :



➤ Appendix 4 – Airbus A320Neo range calculus validation :

